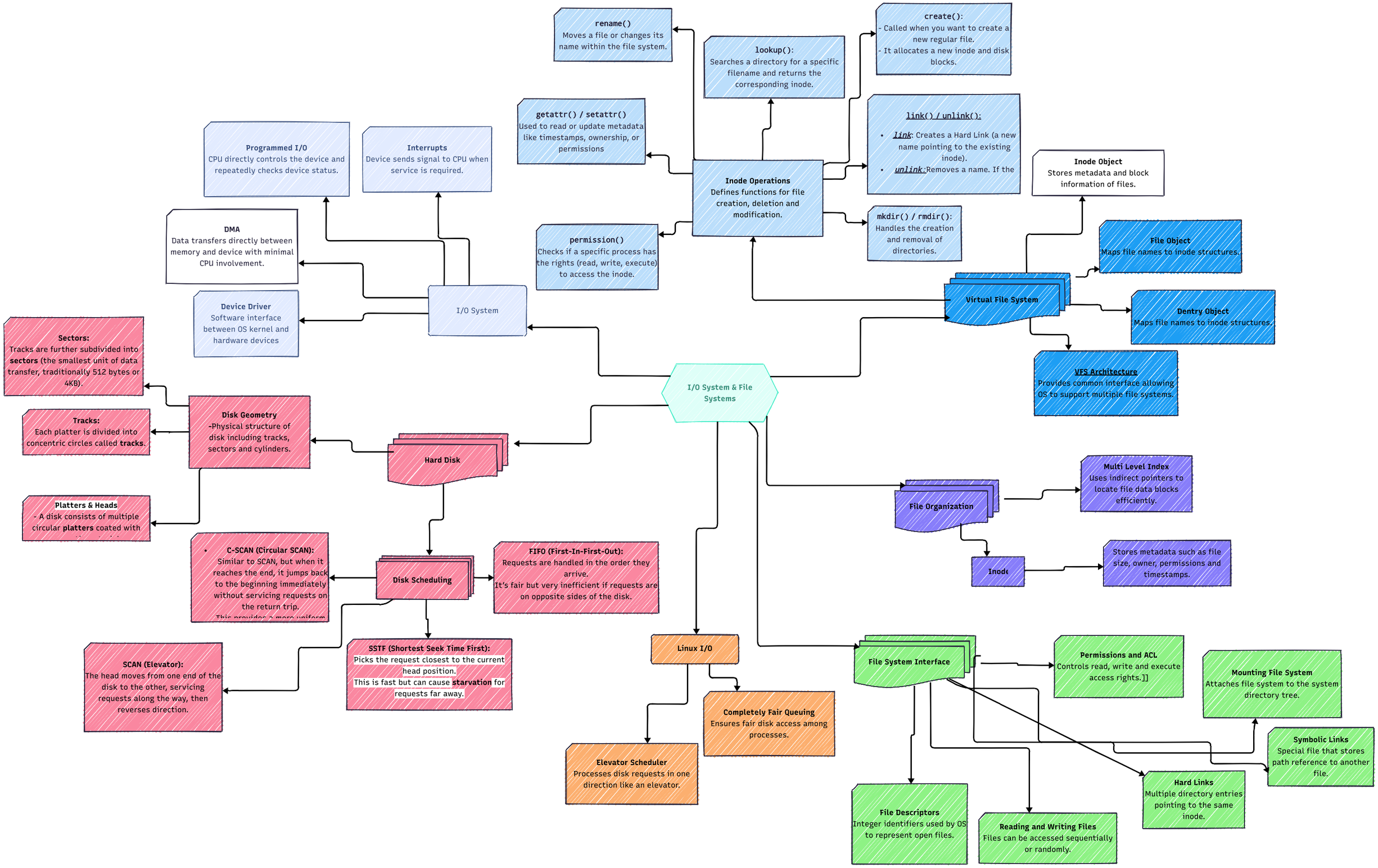
rename()
Moves a file or changes its name within the file system.
lookup():
Searches a directory for a specific filename and returns the corresponding inode.
create():
- Called when you want to create a new regular file.
- It allocates a new inode and disk blocks.
getattr() / setattr()
Used to read or update metadata like timestamps, ownership, or permissions
link() / unlink():
link: Creates a Hard Link (a new name pointing to the existing inode).
unlink: Removes a name. If the
Programmed I/O
CPU directly controls the device and repeatedly checks device status.
Interrupts
Device sends signal to CPU when service is required.
Inode Operations
Defines functions for file creation, deletion and modification.
Inode Object
Stores metadata and block information of files.
DMA
Data transfers directly between memory and device with minimal CPU involvement.
permission()
Checks if a specific process has the rights (read, write, execute) to access the inode.
mkdir() / rmdir():
Handles the creation and removal of directories.
File Object
Maps file names to inode structures.
Device Driver
Software interface between OS kernel and hardware devices
I/O System
Virtual File System
Dentry Object
Maps file names to inode structures.
Sectors:
Tracks are further subdivided into sectors (the smallest unit of data transfer, traditionally 512 bytes or 4KB).
VFS Architecture
Provides common interface allowing OS to support multiple file systems.
I/O System & File Systems
Tracks:
Each platter is divided into concentric circles called tracks.
Disk Geometry
-Physical structure of disk including tracks, sectors and cylinders.
Hard Disk
Multi Level Index
Uses indirect pointers to locate file data blocks efficiently.
File Organization
Platters & Heads
- A disk consists of multiple circular platters coated with
C-SCAN (Circular SCAN):
Similar to SCAN, but when it reaches the end, it jumps back to the beginning immediately without servicing requests on the return trip.
Disk Scheduling
FIFO (First-In-First-Out):
Requests are handled in the order they arrive.
It's fair but very inefficient if requests are on opposite sides of the disk.
Inode
Stores metadata such as file size, owner, permissions and timestamps.
SCAN (Elevator):
The head moves from one end of the disk to the other, servicing requests along the way, then reverses direction.
SSTF (Shortest Seek Time First):
Picks the request closest to the current head position.
This is fast but can cause starvation for requests far away.
Linux I/O
File System Interface
Permissions and ACL
Controls read, write and execute access rights.]]
Mounting File System
Attaches file system to the system directory tree.
Completely Fair Queuing
Ensures fair disk access among processes.
Symbolic Links
Special file that stores path reference to another file.
Elevator Scheduler
Processes disk requests in one direction like an elevator.
Hard Links
Multiple directory entries pointing to the same inode.
File Descriptors
Integer identifiers used by OS to represent open files.
Reading and Writing Files
Files can be accessed sequentially or randomly.